Simulation Assignment 2: Virtual Lab Experiment

7B2M0 Building Performance and Energy Systems Simulation

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1 Introduction

This assignment investigates the transient thermal response of an insulation material using a virtual laboratory experiment in DesignBuilder/EnergyPlus. The provided model represents the Building Physics and Services (BPS) laboratory and contains a test chamber divided into two sections, A and B, as shown in Figure 1. Each chamber has dimensions of $1 \times 2 \times 3.5$ m and contains a cooling unit, while the floor and roof are treated as adiabatic surfaces.

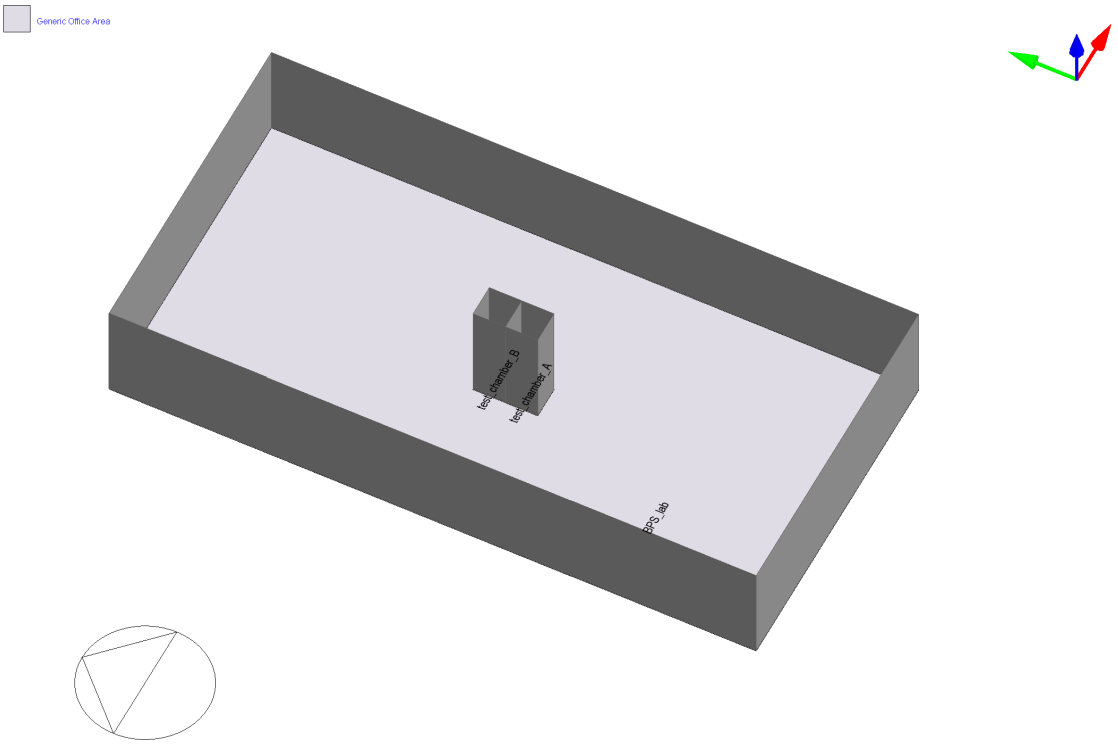


Figure 1: DesignBuilder model of the BPS laboratory and test chambers A and B.

The experiment consists of two main parts. First, the test chambers must be maintained at approximately 0 °C while the surrounding BPS laboratory is maintained at 20 °C. The required thermal transmittance and corresponding wall thickness are therefore determined from the cooling capacity and the heat transfer between the chambers and the surrounding laboratory. The calculated construction is subsequently implemented in DesignBuilder and the resulting initial chamber temperature is verified.

In the second part, a sudden change in the chamber temperature is used to investigate the transient thermal response of the insulation. The simulated inside surface temperature is compared with the analytical transient heat conduction solution provided with the assignment, based on the formulation of Hensen and Nakhi [1]. The simulation is then repeated at different temporal resolutions to investigate how the simulation timestep affects the representation of the rapid transient response.

2 Part 1: Initial State and Wall Thickness Calculation

2.1 Calculating the Required U-Value

The test chamber is maintained at 0 °C while the surrounding BPS lab is maintained at 20 °C. Since the first name "Yusuf" contains five letters, the assigned cooling power is $Q_c = 90$ W for each chamber. With two chambers, the total cooling power is therefore

$$Q_{c,\mathrm{tot}} = 2 \times 90 = 180 \text{ W}. \tag{1}$$

The floor and roof are adiabatic, so heat transfer only occurs through the vertical outer walls of the test chamber. For the chamber dimensions $1 \times 2 \times 3.5$ m, the total heat-transfer area is

$$A = 2(2 \times 3.5) + 2(1 \times 3.5) = 21 \text{ m}^2. \tag{2}$$

At steady state, the heat entering the test chamber through its walls is balanced by the cooling power. Using

$$Q = UA\Delta T, \tag{3}$$

with a temperature difference of

$$\Delta T = 20 - 0 = 20 \text{ K}, \tag{4}$$

the required U-value is

$$U = \frac{Q_{c,\text{tot}}}{A\Delta T} = \frac{180}{21 \times 20} = 0.429 \text{ W/(m}^2\text{K)}. \tag{5}$$

Therefore, the required U-value of the test chamber walls is

$$\boxed{U \approx 0.429 \text{ W/(m}^2\text{K)}}. \tag{6}$$

2.2 Calculating the Required Wall Thickness

The `test_chamber_wall` construction consists of a single insulation layer with an initial thickness of

$$d_{\text{initial}} = 0.080 \text{ m}. \tag{7}$$

For this construction, DesignBuilder reports a surface-to-surface U-value of

$$U_{s-s} = 0.625 \text{ W/(m}^2\text{K)}. \tag{8}$$

Since the surface-to-surface U-value represents the conductive resistance of the wall material, its thermal resistance can be calculated as

$$R_{\text{wall}} = \frac{1}{U_{s-s}} = \frac{1}{0.625} = 1.60 \text{ m}^2\text{K/W}. \tag{9}$$

For a homogeneous material,

$$R_{\text{wall}} = \frac{d_{\text{initial}}}{\lambda}, \tag{10}$$

and therefore the thermal conductivity of the wall material is

$$\lambda = \frac{d_{\text{initial}}}{R_{\text{wall}}} = \frac{0.080}{1.60} = 0.050 \text{ W/(mK)}. \tag{11}$$

The convective heat transfer coefficients on both sides of the chamber wall are fixed in the DesignBuilder model at

$$h_i = h_o = 3 \text{ W/(m}^2\text{K)}. \tag{12}$$

Using the required U-value calculated previously,

$$U = 0.429 \text{ W/(m}^2\text{K)}, \tag{13}$$

the corresponding total thermal resistance is

$$R_{\text{tot}} = \frac{1}{U} = \frac{1}{0.429} \approx 2.333 \text{ m}^2\text{K/W}. \tag{14}$$

For the single-layer chamber wall, the total resistance consists of the convective surface resistances on both sides and the conductive resistance of the insulation:

$$R_{\text{tot}} = \frac{1}{h_o} + \frac{d}{\lambda} + \frac{1}{h_i}. \tag{15}$$

Substituting the known values gives

$$2.333 = \frac{1}{3} + \frac{d}{0.050} + \frac{1}{3}. \tag{16}$$

Solving for the required wall thickness,

$$d = 0.050 \left(2.333 - \frac{1}{3} - \frac{1}{3} \right) \approx 0.0833 \text{ m}. \tag{17}$$

Therefore, the required thickness of the test chamber walls is

$$\boxed{d \approx 0.0833 \text{ m} = 83.3 \text{ mm}}. \tag{18}$$

This thickness is used for the `test_chamber_wall` construction in DesignBuilder for both chambers A and B.

2.3 Simulation and Initial Temperature Verification

The simulation was run in DesignBuilder using EnergyPlus, and the sub-hourly results for test chamber A are shown in Figure 2.

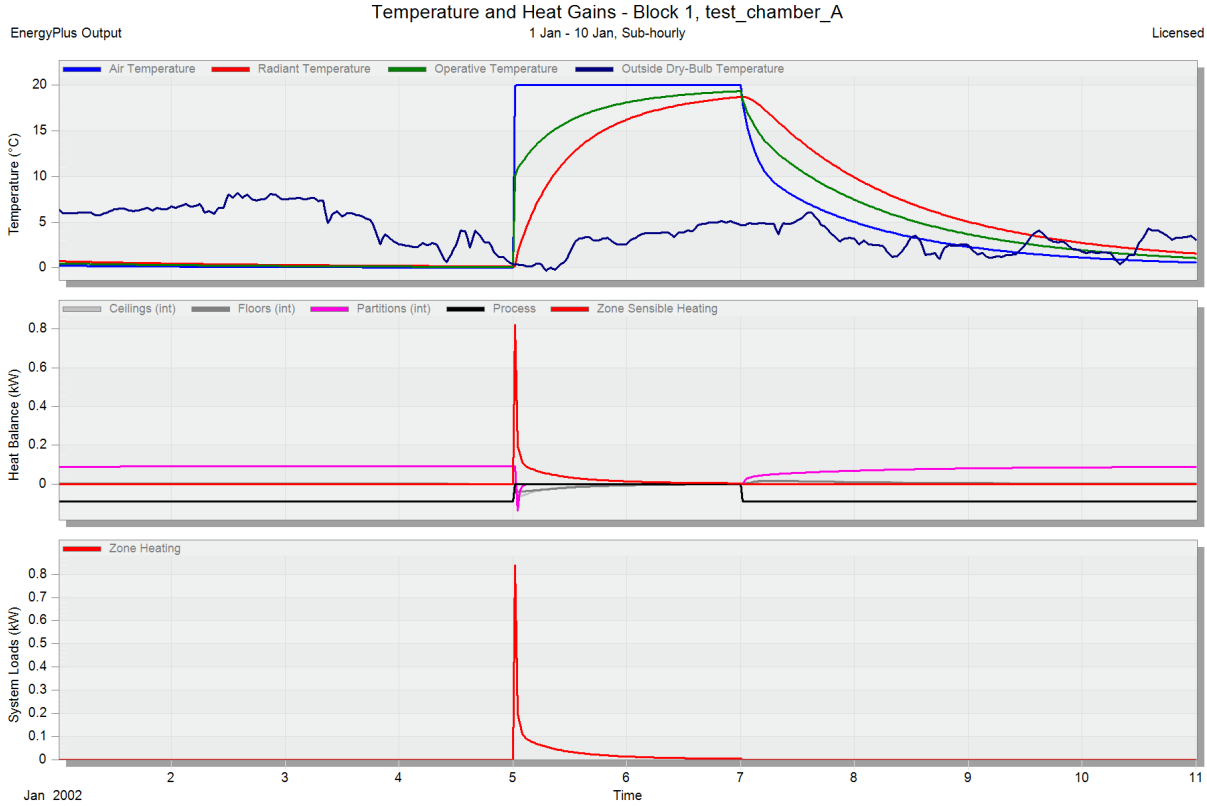


Figure 2: Sub-hourly temperature and heat gain results for test chamber A.

As shown in Figure 2, the air temperature in the test chamber remains approximately 0 °C before the start of the experiment on 5 January. At this point, the cooling is switched off and the heating system rapidly increases the air temperature to approximately 20 °C. The simulation therefore confirms that the required initial chamber temperature of approximately 0 °C is achieved. The results for test chamber B were also checked and showed the same temperature behaviour as chamber A.

3 Part 2: Transient Response and Analysis

The transient response of the insulation material was investigated by applying a sudden temperature change in the test chamber. Initially, the chamber and insulation were maintained at approximately 0 °C. As shown in Figure 2, at the start of the experiment the cooling system was switched off and the heating system was activated, causing the chamber air temperature to rapidly increase towards 20 °C. This temperature step provides the transient condition used to evaluate the thermal response of the insulation material.

The insulation layer has a thickness of 0.10 m, a thermal conductivity of 0.045 W/(m·K), a density of 50 kg/m^3, and a specific heat capacity of 840 J/(kg·K). A convective heat transfer coefficient of 3 W/(m^2·K) was applied on both sides of the construction.

3.1 Comparison with the Analytical Solution

To evaluate the transient behaviour predicted by DesignBuilder/EnergyPlus, the simulated inside surface temperature of the insulation was compared with an analytical solution for transient heat conduction. The analytical solution was provided with the assignment and is based on the formulation presented by Hensen and Nakhi [1].

Figure 3 compares the analytical surface temperature with the corresponding DesignBuilder simulation result. For this initial simulation, a temporal resolution of 30 minutes (0.5 h) was used. Time is expressed relative to the temperature step, such that $t = 0$ represents the start of the temperature change. The DesignBuilder results are therefore available at discrete 30-minute intervals, while the analytical solution provides a much finer representation of the transient response.

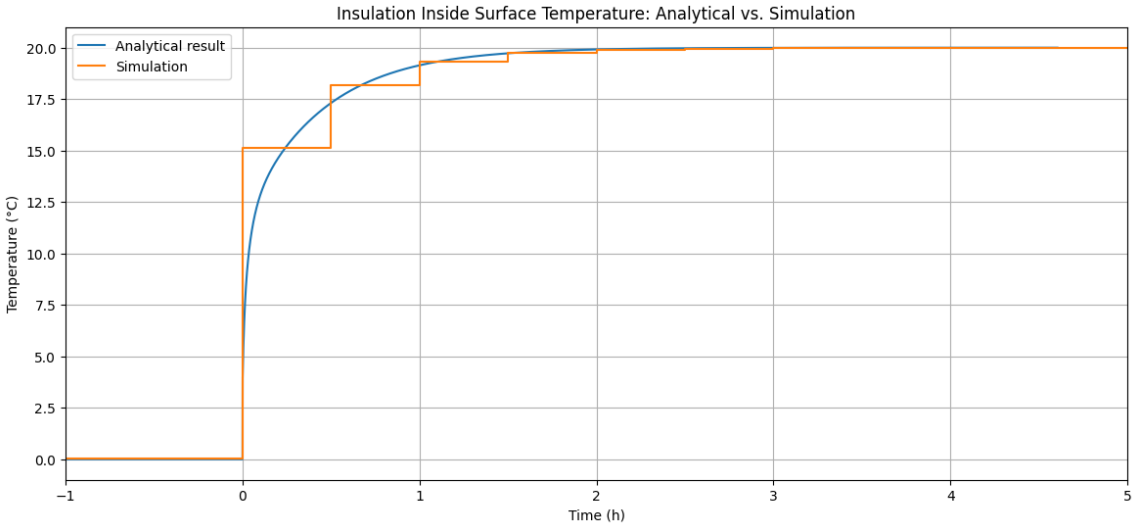


Figure 3: Comparison of the analytical and simulated inside surface temperature of the insulation using a 30-minute simulation time step.

Both results show the same general transient behaviour. Following the temperature step, the inside surface temperature increases rapidly and gradually approaches the steady-state value of approximately 20 °C. The DesignBuilder simulation follows the overall analytical response reasonably closely.

The largest difference is observed during the initial part of the transient. The analytical solution predicts a continuous and rapid increase immediately after $t = 0$, whereas the 30-minute simulation represents the response at discrete time intervals. As time progresses, the difference between the two results decreases, and both approaches converge towards approximately 20 °C.

3.2 Effect of Temporal Resolution

To investigate the influence of temporal resolution on the transient response, the simulation was repeated using four different time-step settings: 2, 4, 12, and 60 timesteps per hour, corresponding to time intervals of 30, 15, 5, and 1 minute, respectively. The resulting inside surface temperatures are compared with the analytical solution in Figure 4.

The results show that temporal resolution has a strong influence on the simulation during the initial transient. With 2 timesteps per hour, the simulated response is represented by relatively large temperature changes between consecutive time steps. As a result, the rapid temperature increase immediately after $t = 0$ is only captured approximately. Increasing the resolution to 4 timesteps per hour reduces these differences, although the discrete nature of the simulated response is still clearly visible.

At 12 timesteps per hour, the simulation follows the analytical curve much more closely, particularly during the first hour after the temperature step. The highest resolution of 60 timesteps per hour provides the closest agreement with the analytical solution and captures the shape of the rapid initial response with considerably more detail.

The influence of temporal resolution becomes less important as the system approaches steady state. After the initial transient, all four simulations converge towards approximately 20 °C and the differences between the time-step settings become very small. This indicates that a coarse time step can reproduce the long-term equilibrium temperature reasonably well, but it may not adequately describe short and rapidly changing thermal processes.

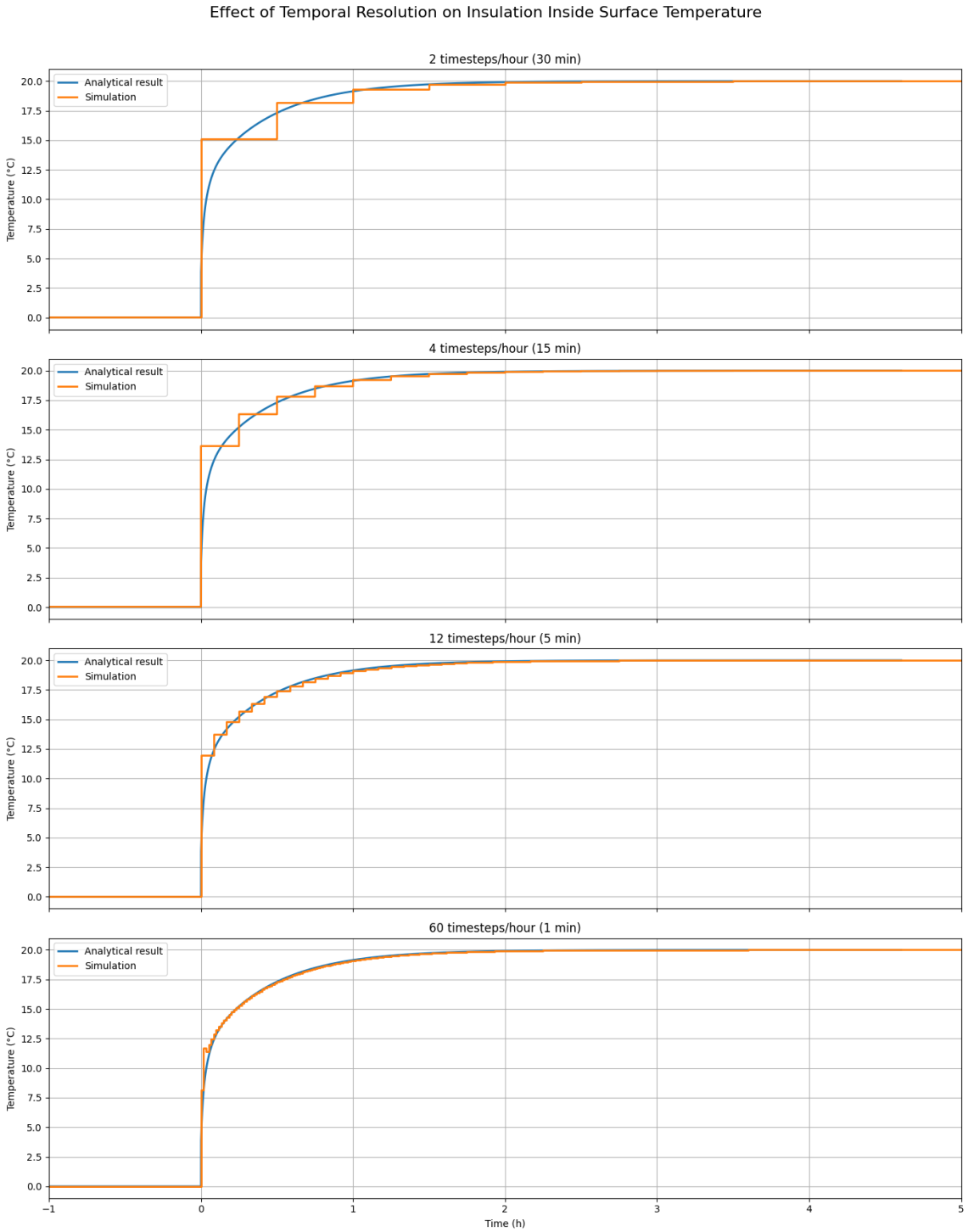


Figure 4: Comparison of the analytical solution and simulated inside surface temperature using different temporal resolutions.

For this case, temporal resolution is therefore particularly important when the objective is to analyse the transient behaviour immediately following a sudden change in boundary conditions. A finer time step provides more calculation points during this period and gives a more detailed representation of the thermal response. However, this improvement comes at the cost of a larger number of calculations. The appropriate time step should therefore balance computational effort with the level of temporal detail required for the analysis.

4 Conclusion

This assignment investigated both the steady-state and transient thermal behaviour of a virtual test chamber using DesignBuilder/EnergyPlus. For the initial condition, the heat balance between the cooled test chambers and the surrounding BPS laboratory resulted in a required U-value of approximately 0.429 W/(m^{2}K). Based on the thermal properties of the chamber wall and the specified surface heat transfer coefficients, a wall thickness of approximately 0.0833 m was obtained. After implementing this value in DesignBuilder, the simulation showed that both chambers could be maintained at approximately 0 °C before the transient experiment.

The transient analysis showed that the simulated insulation surface temperature follows the same overall behaviour as the analytical solution. After the temperature step, the surface temperature rises rapidly before gradually approaching the new steady-state value of approximately 20 °C. The largest discrepancies occur immediately after the temperature change, where the thermal response changes most rapidly.

The temporal-resolution study demonstrates that these initial differences are strongly related to the selected simulation timestep. At 30-minute intervals, the rapid transient is represented by only a small number of discrete values, resulting in relatively large steps and a limited representation of the analytical curve. Reducing the timestep progressively improves the agreement. At 5-minute resolution the transient response is already represented much more closely, while the 1-minute simulation follows the analytical solution very closely throughout most of the response.

An important observation is that the influence of temporal resolution is not equally significant throughout the simulation. Once the system approaches steady state, all investigated timestep settings converge to approximately the same temperature. A coarse temporal resolution may therefore be sufficient if only long-term or steady-state behaviour is of interest. In contrast, when rapid changes in boundary conditions, peak responses, or short-duration thermal processes are important, a finer timestep is required to avoid losing relevant dynamic behaviour.

There are also limitations to the comparison. The analytical solution provides a continuous theoretical reference, whereas the EnergyPlus results are obtained at discrete simulation intervals. Consequently, some difference between the curves, particularly immediately after the temperature step, is inherent to the temporal discretization. Furthermore, the comparison in this assignment is primarily qualitative. A more rigorous assessment could calculate quantitative error measures between the analytical and simulated temperatures for each temporal resolution. This would make it possible to evaluate how much accuracy is gained when reducing the timestep and whether further timestep refinement provides meaningful improvement relative to the additional computational cost.

Overall, the experiment demonstrates that selecting an appropriate temporal resolution is an important part of transient building-performance simulation. The timestep should therefore be chosen according to the physical timescale of the phenomenon being studied rather than solely according to the required simulation duration.

References

[1] J. L. M. Hensen and A. E. Nakhi, “Fourier and biot numbers and the accuracy of conduction modelling,” 1994.